

A spline decomposition of monthly crime trends

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Overview

This paper develops a first-pass seasonal-trend decomposition for monthly crime counts observed across agencies. The motivating idea is the aggregated logit-binomial model used by Wheeler and Kovandzic (Wheeler and Kovandzic 2018), extended so that the time structure is estimated with smooth functions. The goal is descriptive: identify the common curve for each crime type, measure agency departures from that curve, and flag unusually large monthly residuals.

The source snapshot comes from the Real-Time Crime Index repository (AH-Datalytics 2026). The initial fit uses component crimes rather than aggregate violent and property totals, which prevents the same component offenses from entering the stacked model twice.

Data and model

The analysis uses monthly agency-level counts for murder, rape, robbery, assault, burglary, theft, and motor-vehicle theft. Rows are restricted to the published `size == all` sample, the current population threshold, and valid counts. The current fit uses a minimum population of 250,000 and contains 75,813 stacked agency-month-crime observations from 100 agencies.

For agency (*i*), crime type (*c*), and month (*t*), let (Y_{ict}) be the reported count and (N_{ict}) the agency population. The response is

$$Y_{ict} \sim \text{Binomial}(N_{ict}, p_{ict}), \quad \text{logit}(p_{ict}) = \eta_{ict}.$$

The linear predictor is

$$\eta_{ict} = \alpha_c + f_c(\text{time}_t) + g_c(\text{year}_t) + h_c(\text{month}_t) + b_i + b_{ic} + q_{ic}(\text{time}_t) + r_{ict}.$$

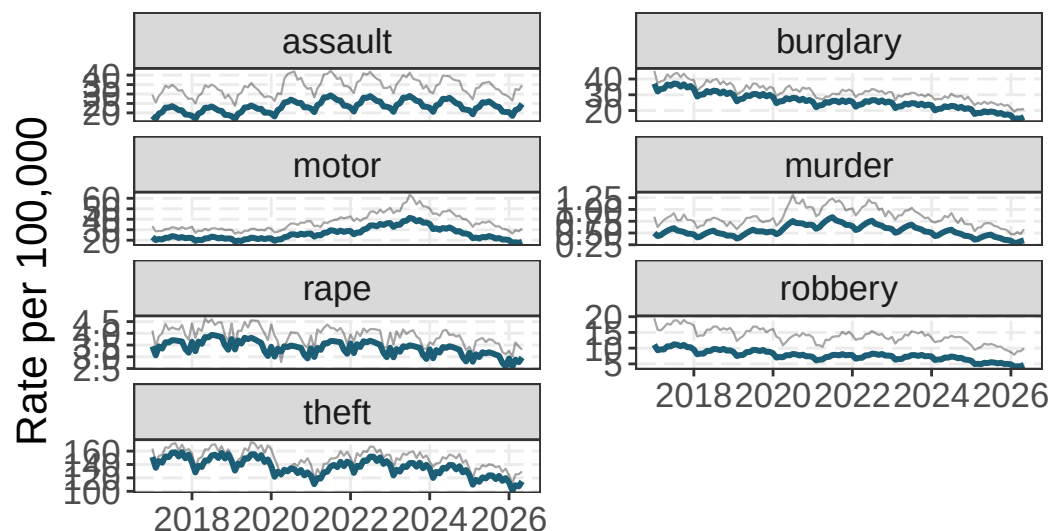
Here (f_c) is the crime-specific global time curve, (g_c) is a smooth year term, and (h_c) is a cyclic cubic spline for month. Agency and agency-by-crime random effects provide partial pooling. The regularized agency-by-crime random time slope (q_{ic}) estimates a persistent departure from the global curve. The agency-month term (r_{ict}), enabled in the primary fit and shared across the stacked crime types, is the overdispersion term analogous to the (`1|Row`) term in the motivating model.

The global baseline is obtained by predicting with agency and observation terms excluded. Agency departures are the difference between the retained agency prediction and this baseline on the logit scale. A monthly observation is flagged when its absolute Pearson residual is at least 3; this threshold is an exploratory screening rule, not a multiple-testing-adjusted hypothesis test.

Global trends

Global monthly trends

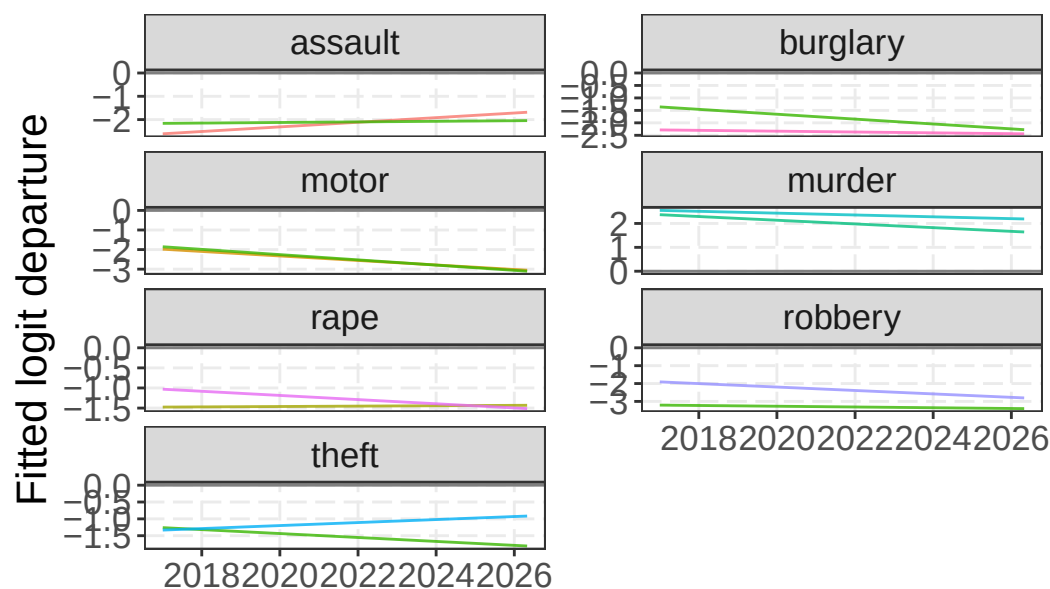
Thin: observed rate; thick: global spline



The thick curves are the population-weighted global spline predictions. The thin curves show the corresponding observed rates. This view separates the slowly evolving crime-specific baseline from month-to-month noise without interpreting every local peak as a structural break.

Agency departures

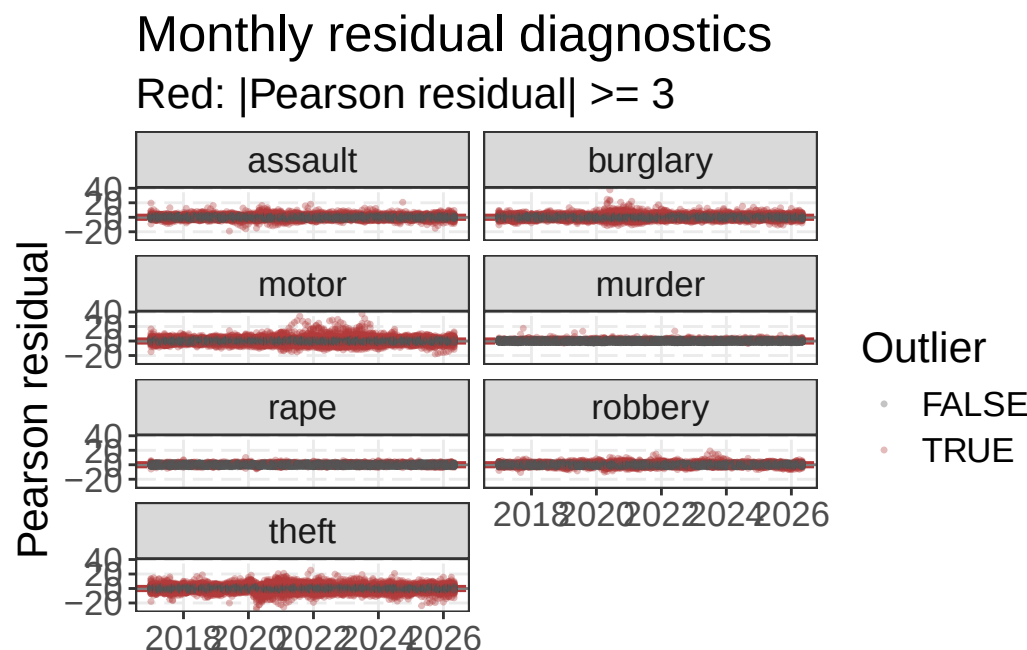
Agency departures from global spline



The figure displays agencies with the largest absolute fitted departures for each crime type. A positive value means that the agency's fitted rate is above the global crime-specific curve on the logit scale. The agency

smooths are regularized, so short isolated spikes are not automatically converted into a new long-run trend.

Residual screening



Residual screening is available at both the month and curve levels. The browser app can filter a selected agency and crime type, change the date window, and change the residual threshold. This makes it possible to inspect whether a flagged month is isolated, part of a run, or aligned with a broader departure in the agency curve.

Limitations and next steps

The source data are reported crime counts and may contain missing reporting, changes in agency coverage, and revisions. The current denominator is the published population field; it is treated as the binomial number of trials for this first implementation. The model is descriptive and does not identify causal effects. In particular, the time spline, year spline, seasonality, and agency smooths should be interpreted together rather than as independent causal components.

Next steps are to add reporting-completeness diagnostics, compare alternative denominators and quasi-binomial formulations, study sensitivity to population thresholds, and automate a pinned monthly update workflow. The supplied Wheeler paper and spline discussion by Simpson provide useful guidance for prediction intervals and comparisons between smooths (Wheeler and Kovandzic 2018; Simpson 2017).

AH-Datalytics. 2026. “Real-Time Crime Index Repository.” <https://github.com/AH-Datalytics/rtci>.

Simpson, Gavin L. 2017. “Difference Splines i.” From the bottom of the heap. <https://fromthebottomoftheheap.net/2017/10/11/difference-splines-i/>.

Wheeler, Andrew P., and Tomislav V. Kovandzic. 2018. “Monitoring Volatile Homicide Trends Across US Cities.” *Homicide Studies* 22 (2): 119–44.